

Task-4: IoT Cloud Integration & Real-Time Remote Monitoring

Objective

Upgrade the Task-3 Smart Automation System into a cloud-connected IoT device that sends live sensor data to the internet and allows remote monitoring via a web dashboard.

Hardware / Simulator

- ESP32 (simulated on Wokwi)
- PIR Motion Sensor
- LED (status indicator)

Cloud Platform

- **ThingSpeak** (free IoT cloud platform)

- Field 1: Motion Status (0 = no motion, 1 = motion detected)

How It Works

1. The PIR sensor connected to ESP32 GPIO 4 continuously monitors for motion.
2. The ESP32 connects to WiFi (Wokwi-GUEST network in simulation).
3. Every 15 seconds, the ESP32 reads the PIR sensor value and sends it to the ThingSpeak cloud via an HTTP GET request to the ThingSpeak Update API, using a unique Write API Key tied to the channel.
4. ThingSpeak stores the value in Field 1 and plots it live on a graph.
5. The onboard LED also mirrors the motion status locally (lights up when motion is detected).

Working Logic

Sensor → Microcontroller (ESP32) →
WiFi → ThingSpeak Cloud → Dashboard

→ User

Files

- `sketch.ino` — Full ESP32 Arduino code (replace `YOUR_WRITE_API_KEY` with your own ThingSpeak Write API Key before running)

Deliverables

- Circuit diagram screenshot (Wokwi)
- ThingSpeak dashboard screenshot (live graph)
- Source code (this repo)
- Explanation (above)

Outcome

A working cloud-connected IoT system demonstrating real-time device-to-cloud communication and remote monitoring — a portfolio-grade mini IoT project.